r/NoStupidQuestions

Ai power consumption vs humans?

Posted by u/toxicatedscientist • • 0 points • 13 comments

So this isn't meant to advocate a position but looking for information: one of the criticisms for Ai content generation is how much power it uses to generate a video or picture or whatever, taking humans jobs, etc. but I'm wondering if it takes the human more or less power to make the same video. Ai is obvs quicker, which is what got me wondering: the human is gonna take longer and go through more iterations, more gpu load, etc. has this actually been compared yet?

Comments

u/LateHippo7183 • 1 points • 2025-12-23 18:05 UTC

As other commentors have said, it's really hard to do a direct comparison.

But it's not like you turn off your computer and go outside as soon as AI has done your task for you. You move on to the next task. So, instead of producing one art piece over 8 hours, you can produce hundreds. Same amount of human labor, but a *lot* more energy consumed.

> **u/Ashamed_Counter_5348** • 1 points • 2025-12-24 03:31 UTC
>
> Are you implying that it takes the same amount of energy for a computer to produce an art piece as a human? Because like, no.
>
> It takes less than 1 watt-hour for a computer to produce a piece of art. It takes one GPU in some big fuckass datacenter about 7 seconds.

u/shadow-battle-crab • 1 points • 2025-12-23 16:26 UTC

The laws of economics still apply. However much energy it takes to create something, you have to pay for - the companies that run these tools aren't burning energy out of the kindness of their hearts, they are here to make a profit. So how much energy is spent in making a thing has to be less than how much money you spent to make it. Lets say 50% of the cost of using AI is raw energy and not company profits, and a chatgpt subscription costs $20 a month, that means you at most use 55 kwh or so of electricity per month.

For comparison an average person's home uses 30 kwh / day, and on hot days this is about 5 hours of running central air AC.

u/Ashamed_Counter_5348 • 1 points • 2025-12-23 03:20 UTC

If it helps, the resting wattage (heat output and therefore energy consumption) of a human being is about 200 watts.

It's a difficult comparison to make, though. Running a human at 200 watts is certainly more expensive than running a robot at 200 watts, cause you know, corn and beef are more expensive per watt than gasoline. It'd be more meaningful to perform a calculation of raw economic cost.

Either way, the robots win. Sorry.

u/toxicatedscientist • 1 points • 2025-12-23 05:11 UTC

I actually want to compare the energy used by the humans computer, not the human. The whole "gaming pc is a space heater" sorta thing

u/Ashamed_Counter_5348 • 1 points • 2025-12-24 03:26 UTC

I don't think there's any way to make an apples-to-apples comparison there. It depends on the type of content, the method of creation, etc. If you're making a video, what kind of video is it? Are you using the output of the AI directly (I.E. tiktok slop channels) or are you using the output as a small part of a bigger video?

As it stands currently, AI is not really capable of automating much except producing a single image or text. Or a weird uncanny video.

When it comes to producing anything people actually want to consume, you can't really just make a prompt and have a finished product. But in any situation where you save time, such as generating a logo, it's probably safe to say you saved energy too.

u/TheApiary • 1 points • 2025-12-22 23:46 UTC

It's gonna depend a lot on how you count. With AI, those numbers are often counting how much energy it took to train it on *everything* not just on what it used to do your specific task. So if you do it for a person, are you gonna count all the energy it took to grow all the food they've eaten for their whole life, all the energy they used for flights and travel, etc?

u/Fast-Appointment-952 • 1 points • 2025-12-26 13:56 UTC

This is exactly why these comparisons are kinda meaningless lol. Like do we count the energy to build the computer the human uses too? What about heating their studio? The coffee they drink while working?

The whole thing gets ridiculous real quick when you try to make it apples to apples

u/toxicatedscientist • 1 points • 2025-12-23 00:31 UTC

I havnt actually personally seen any specific numbers, just useless comparisons like "bottles of water" which aren't applicable? But yea I'm curious about the energy specifically used per task, training is a one time (ish) thing which means that cost gets distributed further as its used more

u/TheApiary • 2 points • 2025-12-23 00:53 UTC

There are a lot of problems with AI, but the good news is, the amount of water you use by doing one additional prompt is simply not one of them. Here's a good roundup: https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about

The best info that we have is that each prompt uses about 2 mL. Which is about the amount your shower uses in 1/40 of a second. So if you want to do 40 AI prompts per day and not use any more water than you used before, end your shower one second earlier. But if you don't think people should spend a lot of effort thinking about how many seconds long their shower is, then it also doesn't make sense to be thinking about the individual impact on water use from your AI prompts

u/toxicatedscientist • 1 points • 2025-12-23 01:25 UTC

… u/bot-sleuth-bot help me out here, is apiary a human?

u/TheApiary • 1 points • 2025-12-23 01:33 UTC

I've been posting on this account for like 10 years, bots couldn't write real comments then

u/toxicatedscientist • 1 points • 2025-12-23 01:34 UTC

Hm. Ok well yea not worried about water

u/TheApiary • 1 points • 2025-12-23 01:36 UTC

To be clear I am worried about some aspects of global water use! Just mostly from animal agriculture and other things that use way more water than chatgpt